

Observation of the suppressed $\Lambda_b^0 \rightarrow [K^+\pi^-]_D p K^-$ decay and measurement of its CP asymmetry

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Analysis twiki page:

<https://twiki.cern.ch/twiki/bin/view/LHCbPhysics/Lb2DpKADS>

Latest ANA (v6.1):

https://twiki.cern.ch/twiki/pub/LHCbPhysics/Lb2DpKADS/main_v61.pdf

RC e-group: lhcb-review-B2OC-Lb2DpK@cern.ch

RC: Steve Blusk (chair), Fabio Ferrari

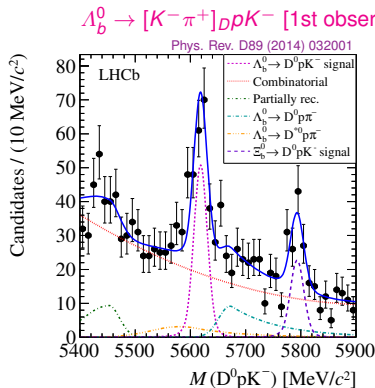
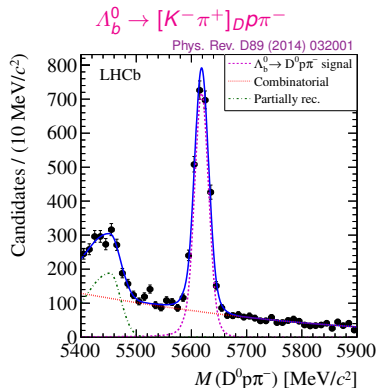
EB reviewer: Franz Muheim

B2OC WG reader: Nicole Skidmore

Analysis approval
22 Jun 2021

Introduction

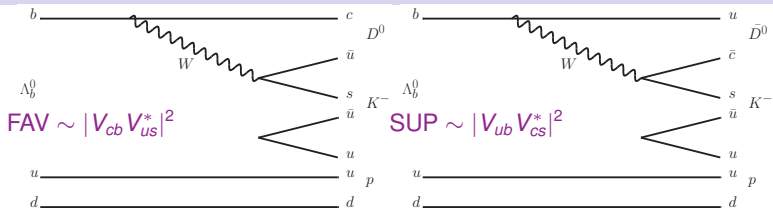
- Few studies of beauty baryon decays to final states involving a single open-charm meson exist



- In this analysis we study even further suppressed $\Lambda_b^0 \rightarrow [K^+ \pi^-]_D p K^-$ decay:

- $\Lambda_b^0 \rightarrow [K^- \pi^+]_D p K^- \sim |V_{cb} V_{us}^*|^2$ is relatively favoured (FAV)
 - $\Lambda_b^0 \rightarrow [K^+ \pi^-]_D p K^- \sim |V_{ub} V_{cs}^*|^2$ is relatively suppressed (SUP): *not yet observed*
- (FAV and SUP are defined by the relative charges of K from D and K from Λ_b^0)

Analysis goals and strategy



- $R_{\text{pred}} = \left| \frac{V_{cb} V_{us}^*}{V_{ub} V_{cs}^*} \right|^2 \approx 7.4$
- The suppressed decay receives contributions from $\Lambda_b^0 \rightarrow [K^+ \pi^-]_{D^0} p K^-$ and $\Lambda_b^0 \rightarrow [K^+ \pi^-]_{\bar{D}^0} p K^-$, leading to interference between them, CP violation and γ sensitivity [ADS method, PRL 78 (1997) 3257, PRD 63 (2001) 036005]
- Goals of this analysis:
 - ▶ observation of suppressed decay $\Lambda_b^0 \rightarrow [K^+ \pi^-]_{D^0} p K^-$
 - ▶ measurement of ratio $R = \frac{\mathcal{B}(\Lambda_b^0 \rightarrow [K^- \pi^+]_{D^0} p K^-)}{\mathcal{B}(\Lambda_b^0 \rightarrow [K^+ \pi^-]_{D^0} p K^-)}$
 - ▶ measurement of asymmetry $A = \frac{\mathcal{B}(\Lambda_b^0 \rightarrow [K^+ \pi^-]_{D^0} p K^-) - \mathcal{B}(\bar{\Lambda}_b^0 \rightarrow [K^- \pi^+]_{\bar{D}^0} \bar{p} K^+)}{\mathcal{B}(\Lambda_b^0 \rightarrow [K^+ \pi^-]_{D^0} p K^-) + \mathcal{B}(\bar{\Lambda}_b^0 \rightarrow [K^- \pi^+]_{\bar{D}^0} \bar{p} K^+)}$
 - ▶ measure R and A in restricted region $M^2(p K^-) < 5 \text{ GeV}^2$ with enhanced γ sensitivity
- Similar to previous analysis:
 - ▶ study of $\Lambda_b^0 \rightarrow D \text{ph}$, $\Lambda_b^0 \rightarrow \Lambda_c h$ [Phys. Rev. D89 (2014) 032001, LHCb-ANA-2012-096]
 - ▶ AmAn $\Lambda_b^0 \rightarrow D p \pi^-$ [JHEP 1705 (2017) 030, LHCb-ANA-2015-072]
- All analysis ingredients (selection, fits etc.) developed without using suppressed data events

Data (Run 1 and 2, 9 fb⁻¹) and MC samples

Year	Stripping version
2011	21r1
2012	21
2015	24r1
2016	28r1
2017	29r2
2018	34

Year	sim. version (signal)
2011–2012	Sim08h
2015–2016	Sim09c
2017	Sim09e
2018	Sim09f

stripping line Lb2D0PHD02HHBeauty2CharmLine

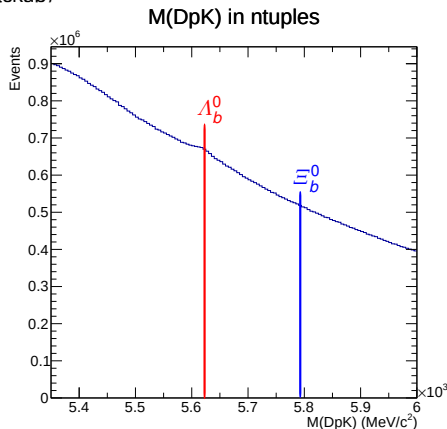
Tracks	χ^2/dof	< 4
	p_T	> 100 MeV/c
	p	> 1000 MeV/c
	IP χ^2	> 4
D^0	Sum of p_T of tracks	> 1800 MeV/c
	DOCA between tracks	< 0.5 mm
	Vertex χ^2/dof	< 10
	IP χ^2 to the best PV	> 36
	Invariant mass	$ M - 1864.84 < 100 \text{ MeV}$
Λ_b^0	Sum of p_T for decay products	> 5000 MeV/c
	Vertex χ^2/dof	< 10
	IP χ^2 to the best PV	< 25
	Lifetime w.r.t best PV	> 0.2 ps
	cos(direction angle) w.r.t best PV	> 0.999
	Invariant mass	$5200 < M < 7000 \text{ MeV}$
PID	Proton from Λ_b^0	$\Delta LL_{p\pi} > -10$
	Kaon from Λ_b^0	$\Delta LL_{K\pi} > -10$
	Kaon from D^0 (only in 2018)	$\Delta LL_{K\pi} > -10$
	Pion from D^0 (only in 2018)	$\Delta LL_{K\pi} < 20$

Simulated sample

	BDT	eff.	cor.	fit	misID	double misID
$\Lambda_b^0 \rightarrow D^0 p K^-, D^0 \rightarrow K^- \pi^+$ (signal)	×	×	×			×
$\Lambda_b^0 \rightarrow D^0 p K^-, D^0 \rightarrow K^- K^+$					×	
$\Lambda_b^0 \rightarrow D^0 p K^-, D^0 \rightarrow \pi^- \pi^+$					×	
$\Lambda_b^0 \rightarrow D^0 p \pi^-, D^0 \rightarrow K^- \pi^+$				×		
$\Lambda_b^0 \rightarrow (D^{*0} \rightarrow [D^0 \rightarrow K^- \pi^+] \gamma(\pi^0)) p K^-$				×		
$\Lambda_b^0 \rightarrow (D^{*0} \rightarrow [D^0 \rightarrow K^- \pi^+] \gamma(\pi^0)) p \pi^-$				×		

Event reconstruction and selection: ntuples

- Similar to previous analyses:
 - ▶ study of $\Lambda_b^0 \rightarrow Dph$, $\Lambda_b^0 \rightarrow \Lambda_c h$ [Phys. Rev. D89 (2014) 032001, LHCb-ANA-2012-096]
 - ▶ AmAn $\Lambda_b^0 \rightarrow Dp\pi^-$ [JHEP 1705 (2017) 030, LHCb-ANA-2015-072]
- DecayTreeFit (DTF) with vertices and D^0 mass constraints
- PID corrections for simulation using `PIDCalib`
- Scaling of topological variables in simulation to match data following LHCb-ANA-2015-072 (distributions in backup)



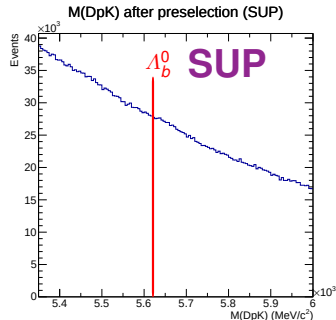
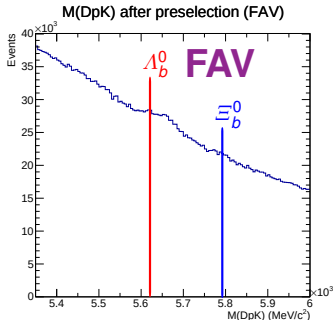
Further event selection ('preselection', or 'pre-TMVA')

Further cleaning of events:

- L0: L0hadron TOS or global TIS
- HLT1 (implicit by HLT2): `Hlt1TrackDecision` in Run1, `Hlt1 (Track|TwoTrack)MVADecision` in Run2 (TOS), or `Hlt1 (DiMuon|SingleMuon|Track)` in both Run1 and Run2 (TIS)
- HLT2: `Hlt2Topo (2|3|4)BodyDecision` in Run2 and `Hlt2 (Topo2Body|Topo3Body|Topo4Body|Charm2Body)BBDTDecision` in Run1
- tracks are not muons
- Λ_c veto (backup)

To fight combinatorial background before training BDT:

- χ^2/dof of DTF < 10
- $c\tau$ (decay time significance) of $\Lambda_b^0 > 0$, $c\tau$ of $D^0 > 0$



BDT selection (TMVA)

- Using ROOT TMVA BDT (adaptive boosting with default settings), 16 variables (distributions in backup):

Variable	Δ_b^0	D^0	π_D	K_D	p	K
χ^2 of DTF	×					
χ^2 of vertex	×	×				
p_T asymmetry in cone of 1.5	×					
$c\tau$ significance	×	×				
IP χ^2	×	×				
$\Delta L_{LK\pi}$			×	×		×
proNN					×	
p_T	×	×	×	×		

► checked also MLP method: similar results

► checked separate training in Run1 and Run2: similar results

- Training samples:

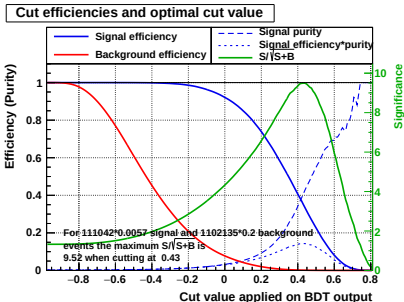
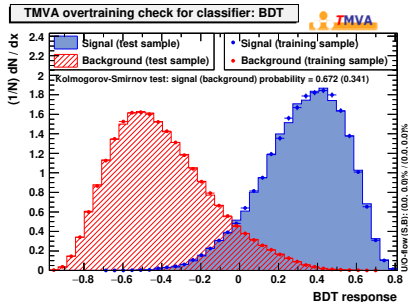
► signal: MC $\Lambda_b^0 \rightarrow (K^- \pi^+) D^0 p K^-$

► background: DATA Λ_b^0 mass sidebands $5300 < M < 5400 \parallel 5900 < M < 6000$ MeV

- Choosing working point (cut on BDT classifier):

► optimise significance ($S/\sqrt{S+B}$) for suppressed decay, assuming $R = \text{FAV}/\text{SUP} = 7.4$

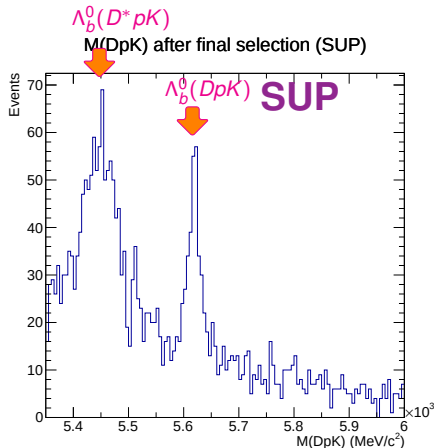
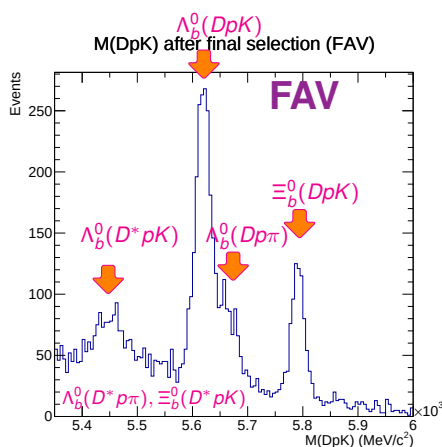
→ expecting suppressed signal $\sim 200 \pm 20$ ($S/\sqrt{S+B} \sim 10$, or 10% stat. unc.)



Final selection

Further cuts applied to suppress charmless background $\Lambda_b^0 \rightarrow ph^- h^+ h^-$:

- significance of D decay time > 2.5
- $|M([K\pi]_D) - m(D)| < 15 \text{ MeV}$

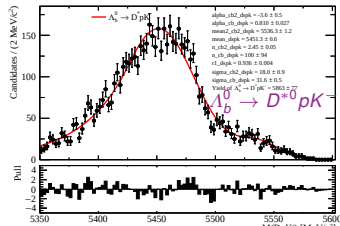
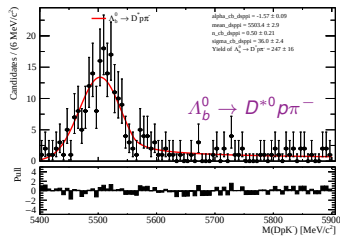
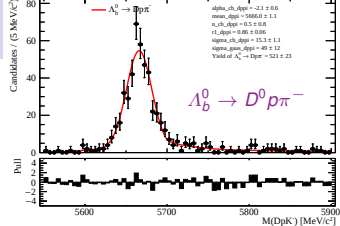
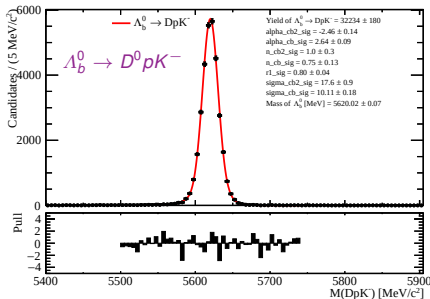


No $\Xi_b^0 \rightarrow DpK(\pi)$, $\Lambda_b^0 \rightarrow D^{(*)}p\pi$ expected in the suppressed decay (totally negligible)

Invariant mass fit: shapes of components

Fit component	FAV	SUP	nominal function
$\Lambda_b^0 \rightarrow D^0 p K^-$	×	×	CB + CB
$\Lambda_b^0 \rightarrow D^0 p \pi^-$	×		Gaus + CB
$\Lambda_b^0 \rightarrow D^{*0} p K^-$	×	×	CB + CB, $m_1 \neq m_2$
$\Lambda_b^0 \rightarrow D^{*0} p \pi^-$	×		CB
$\Xi_b^0 \rightarrow D^0 p K^-$	×		same as for Λ_b^0
$\Xi_b^0 \rightarrow D^{*0} p K^-$	×		same as for Λ_b^0
comb. bkg.	×	×	exponential

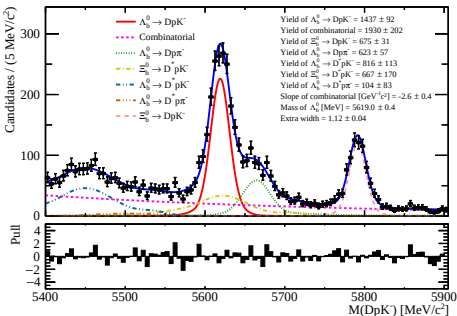
- Ξ_b shapes are taken from Λ_b^0 sim. and shifted by $m(\Lambda_b^0) - m(\Xi_b)$
- Λ_b^0 mass is fitted
- all widths are scaled with extra common parameter free to float



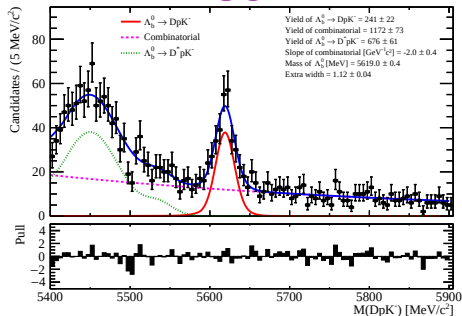
Invariant mass fit (plots for paper w/o pulls in backup)

- Unbinned ML fit (RooFit) performed simultaneously for favoured and suppressed decays, sharing all parameters except yields and combinatorial background slope
- no $\Xi_b^0 \rightarrow D\rho K(\pi)$ and $\Lambda_b^0 \rightarrow D^* \rho \pi$ in suppressed decay (totally negligible)

FAV

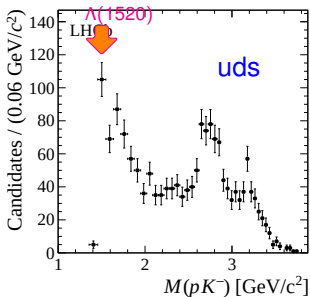
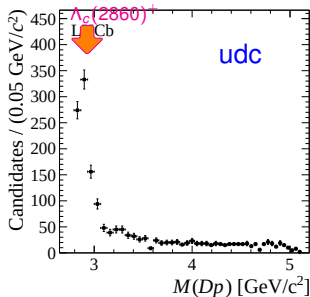
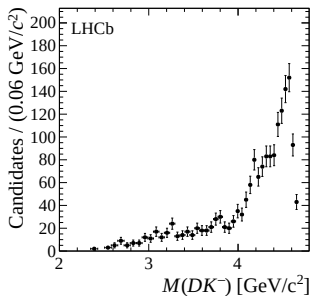
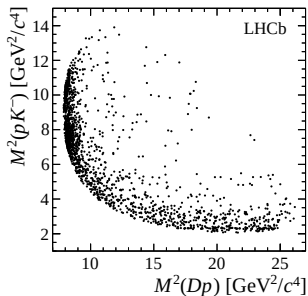


SUP

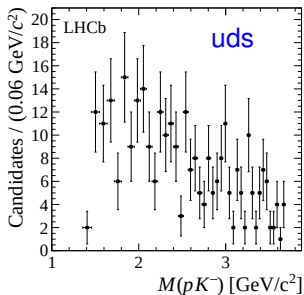
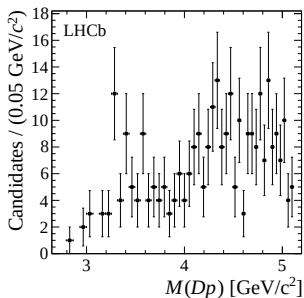
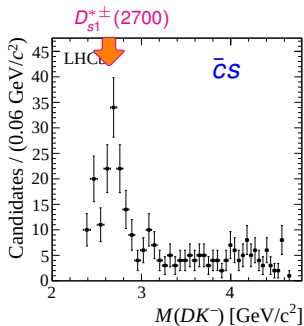
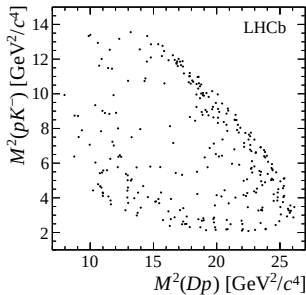


- $N_{\text{FAV}} = 1437 \pm 92$, $N_{\text{SUP}} = 241 \pm 22$ consistent with TMVA training expectation
 - first observation of suppressed decay $\Lambda_b^0 \rightarrow (K^+ \pi^-) D\rho K^-$
- Fit uncertainties cross checked using toys (backup)

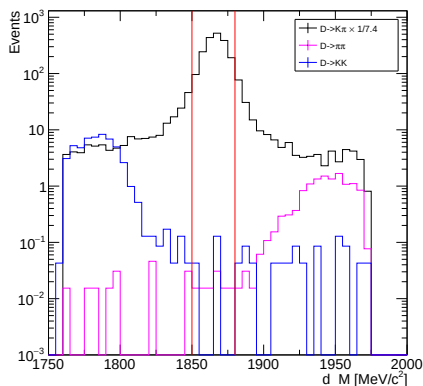
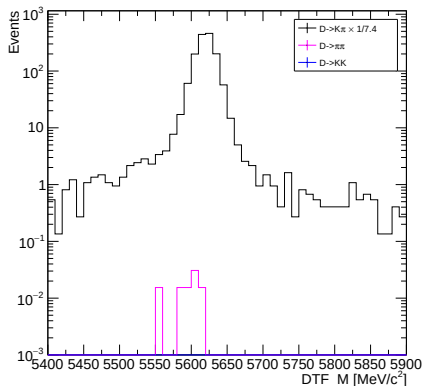
Dalitz plot: favoured decay



Dalitz plot: suppressed decay



Single misID background from $D \rightarrow K^+K^-$, $D \rightarrow \pi^+\pi^-$: $< 0.1\%$

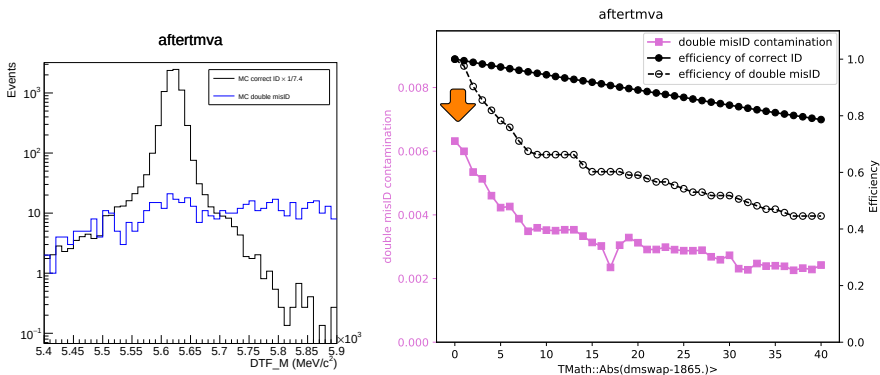


Estimated from MC: **0.01%** and **0.1%** in FAV and SUP, respectively (systematic uncertainty)

- suppressed by ratios $(\mathcal{B}(D^0 \rightarrow K^+K^-) + \mathcal{B}(D^0 \rightarrow \pi^+\pi^-))/\mathcal{B}(D^0 \rightarrow K^-\pi^+) \approx 0.1$
- suppressed by PID variables in BDT
- very strongly suppressed by $|M([K\pi]_D) - m(D)| < 15 \text{ MeV}$

Double misID $K \leftrightarrow \pi$ background (in suppressed decay): $< 1\%$

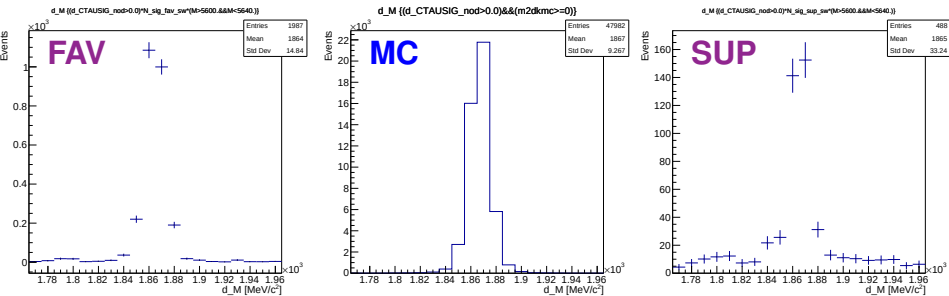
Contamination of SUP mode $\Lambda_b^0 \rightarrow (K^+\pi^-)_D p K^-$ by FAV decay $\Lambda_b^0 \rightarrow (K^-\pi^+)_D p K^-$ with swapped $K \leftrightarrow \pi$ from D is estimated from MC to be 0.6% in SUP (systematic uncertainty):



- Double misID SUP $\rightarrow$ FAV is totally negligible

Charmless Λ_b^0 background

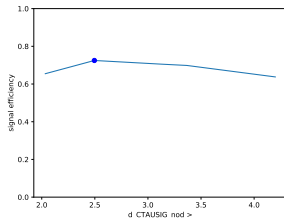
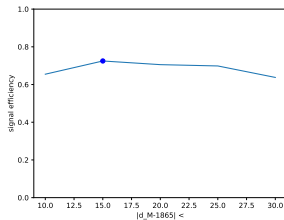
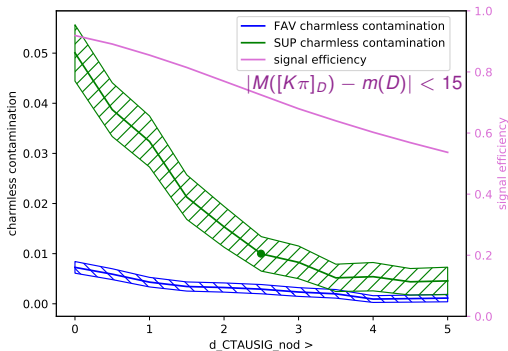
- it is expected to affect SUP more than FAV because of $\Lambda_b^0 \rightarrow pK^-K^+\pi^-$ decay
- can be controlled by checking D invariant mass (d_M) and $c\tau(D^0)$ ($d_CTAUSIG_nod$)
- use sWeights to plot signal(+charmless) as function of d_M



- sidebands in FAV nearly 0
- sidebands in SUP above 0 visibly (but also include small contribution from double misID)
- charmless fraction is estimated by integrating sidebands $50 < |M([K\pi]_D) - m(D)| < 100$ MeV and subtracting (small) $\Lambda_b^0 \rightarrow D^0 pK$ signal fraction estimated using MC

Charmless Λ_b^0 background: $\lesssim 1\%$

- estimate charmless contamination as function of $|M([K\pi]_D) - m(D)|$ and decay time significance (d_CTAUSIG_nod), assuming that charmless component is flat in $M([K\pi]_D)$
- find cuts on $|M([K\pi]_D) - m(D)|$ and d_CTAUSIG_nod to have it $\leq 1\%$ and max. sig. eff.



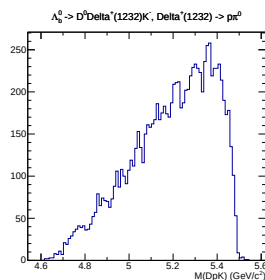
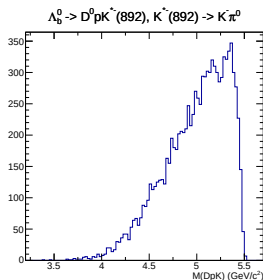
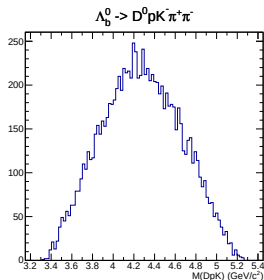
Cuts $|M([K\pi]_D) - m(D)| < 15$ MeV, d_CTAUSIG_nod > 2.5 :

- signal efficiency is 75%
- charmless fraction is 1% in SUP and 0.5% in FAV (sys. unc.)

Other potential backgrounds checked using RapidSim

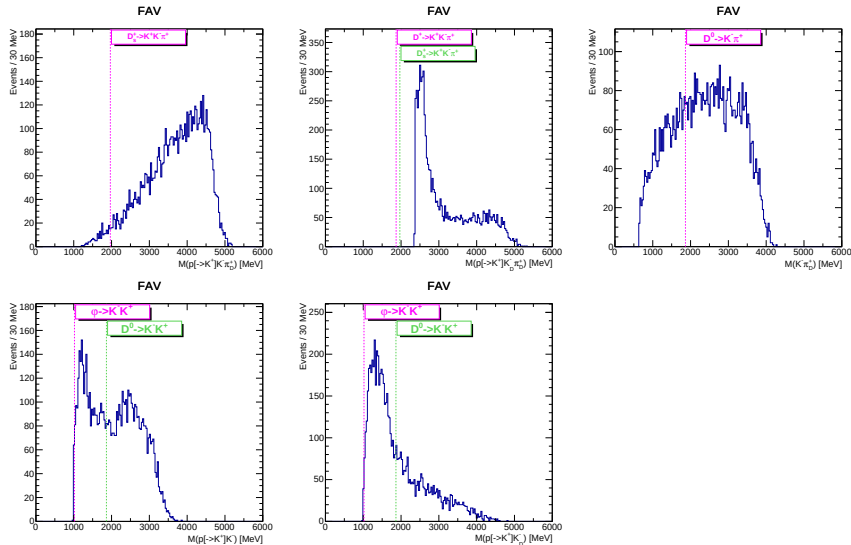
Several decays checked using RapidSim:

- $\Lambda_b^0 \rightarrow DpK^-\pi^+\pi^-$
 - $\Lambda_b^0 \rightarrow D^0pK^{*-}(892), K^{*-}(892) \rightarrow K^-\pi^0$
 - $\Lambda_b^0 \rightarrow D\Delta^+(1232)K^-, \Delta^+(1232) \rightarrow p\pi^0$
- outside of Λ_b^0 peak



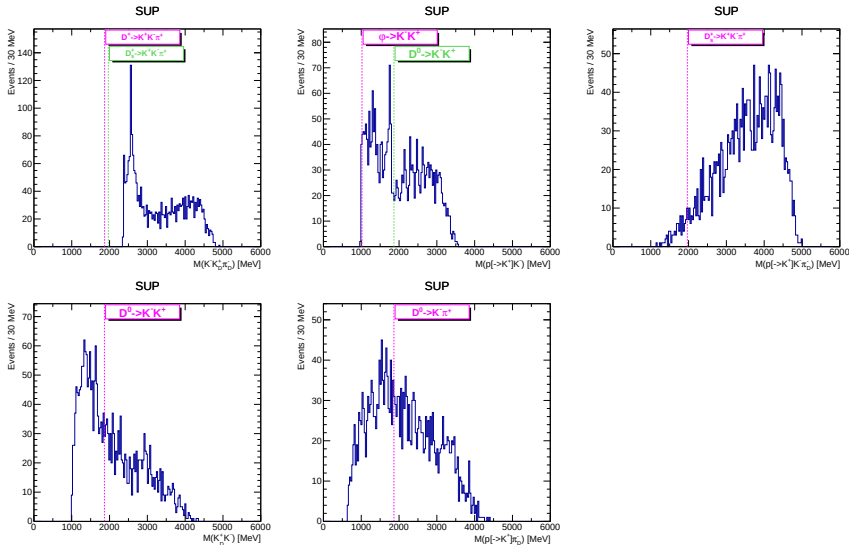
Other potential backgrounds checked in data (FAV)

Further checks for background from $D^+ \rightarrow K^- K^+ \pi^+$, $D_s^+ \rightarrow K^- K^+ \pi^+$, $D^0 \rightarrow K^- \pi^+$, $\phi \rightarrow K^- K^+$ done by plotting inv. mass of various combinations of final-state particles in data:



Other potential backgrounds checked in data (SUP)

Further checks for background from $D^+ \rightarrow K^- K^+ \pi^+$, $D_s^+ \rightarrow K^- K^+ \pi^+$, $D^0 \rightarrow K^- \pi^+$, $\phi \rightarrow K^- K^+$ done by plotting inv. mass of various combinations of final-state particles in data:

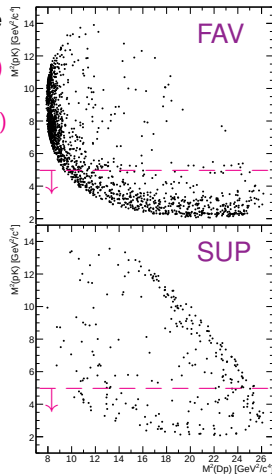
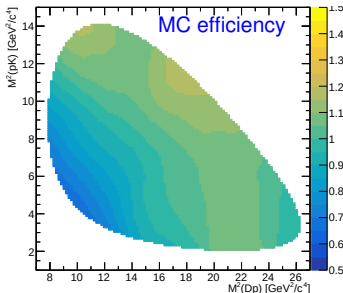


Efficiency correction and measurement of R and A in full phase space

- Efficiency includes reconstruction, selection and kinematic acceptance
- Relative efficiency is determined across the Dalitz plot variables (with D^0 and Λ_b^0 masses constrained), assuming unpolarised Λ_b^0 , and parametrised using kernel-based PDF (Meerkat)
 - ▶ overall, relative efficiency is flat, varying within 0.7–1.2
- s-weights from invariant mass fits and efficiency corrections are

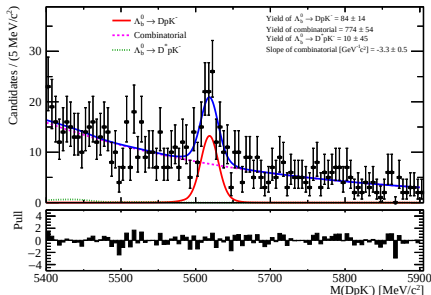
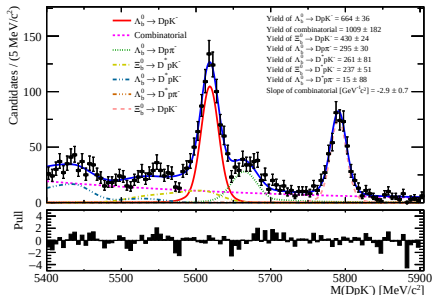
$$\text{▶ } R = \frac{\mathcal{B}(\Lambda_b^0 \rightarrow [K^- \pi^+]_D \rho K^-)}{\mathcal{B}(\Lambda_b^0 \rightarrow [K^+ \pi^-]_D \rho K^-)} = 7.14 \pm 0.79(\text{stat.})_{-0.33}^{+0.43}(\text{syst.})$$

$$\text{▶ } A = \frac{N_{\text{SUP}}(\Lambda_b^0) - N_{\text{SUP}}(\overline{\Lambda_b^0})}{N_{\text{SUP}}(\Lambda_b^0) + N_{\text{SUP}}(\overline{\Lambda_b^0})} = 0.119 \pm 0.088(\text{stat.})_{-0.026}^{+0.024}(\text{syst.})$$



Measurement of R and A in restricted phase space $M^2(pK) < 5 \text{ GeV}^2$

- sensitivity to CP-violation requires interference between amplitudes involving intermediate D^0 and $\bar{D}^0$ mesons in a region of the Dalitz plot
- $M^2(pK) < 5 \text{ GeV}^2$ is chosen which contains the observed $\Lambda(1520)$ contribution, and balances sensitivity and statistics
- the same measurement procedure is used, except
 - ▶ BDT is trained in the $M^2(pK) < 5 \text{ GeV}^2$ region
 - ▶ another parametrised PDF (bifurcated Gaus + CB) for $\Lambda_b^0 \rightarrow D^* pK$ is used (backup)



$$R = 8.55 \pm 1.47(\text{stat.})^{+0.38}_{-0.33}(\text{syst.})$$

$$A = 0.01 \pm 0.16(\text{stat.})^{+0.026}_{-0.024}(\text{syst.})$$

Systematic uncertainties due to fit model

- Variation of parametrised shapes (1–5%):

Fit component	FAV	SUP	nominal function	variation
$\Lambda_b^0 \rightarrow D^0 p K^-$	×	×	CB + CB	Gaus + Gaus
$\Lambda_b^0 \rightarrow D^0 p \pi^-$	×		Gaus + CB	CB + CB
$\Lambda_b^0 \rightarrow D^{*0} p K^-$	×	×	(full) CB + CB, $m_1 \neq m_2$	CB + CB, $m_1 = m_2$
			(restr.) bif. Gaus + CB, $m_1 \neq m_2$	CB + CB, $m_1 \neq m_2$
$\Lambda_b^0 \rightarrow D^{*0} p \pi^-$	×		CB	CB + Gaus
$\Xi_b^0 \rightarrow D^0 p K^-$	×		same as for Λ_b^0	
$\Xi_b^0 \rightarrow D^{*0} p K^-$	×		same as for Λ_b^0	
comb. bkg.	×	×	exponential	1st, 2nd order pol.

- Further cross checks:

- ▶ using signal MC w/o true ro reco matching
- ▶ fixed extra width factor to 1
- ▶ variation of fit range: $5550 < M < 5900$ MeV

Systematic uncertainties due to fit model

	R	A
Gaus+Gaus for $\Lambda_b^0 \rightarrow [K^- \pi^+]_D p K^-$	-0.01	-0.002
CB+CB for $\Lambda_b^0 \rightarrow [K^- \pi^+]_D p \pi^-$	+0.03	+0.000
CB+Gaus for $\Lambda_b^0 \rightarrow [K^- \pi^+]_{D^*} p \pi^-$	-0.00	-0.000
CB+CB with shared mean for $\Lambda_b^0 \rightarrow [K^- \pi^+]_{D^*} p K^-$	+0.37	+0.000
1st order polynomial for combinatorial background	-0.15	-0.010
2nd order polynomial for combinatorial background	-0.15	-0.011
Total variation due to fit model	+0.37 -0.15	+0.000 -0.011

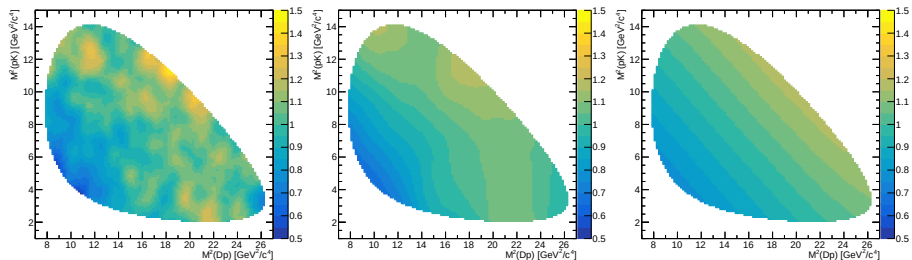
	$R_{M^2(pK^-) < 5 \text{ GeV}^2/c^4}$	$A_{M^2(pK^-) < 5 \text{ GeV}^2/c^4}$
Gaus+Gaus for $\Lambda_b^0 \rightarrow [K^- \pi^+]_D p K^-$	+0.05	+0.011
CB+CB for $\Lambda_b^0 \rightarrow [K^- \pi^+]_D p \pi^-$	+0.01	-0.003
CB+Gaus for $\Lambda_b^0 \rightarrow [K^- \pi^+]_{D^*} p \pi^-$	-0.02	-0.000
CB+CB for $\Lambda_b^0 \rightarrow [K^- \pi^+]_{D^*} p K^-$	+0.02	-0.007
1st order polynomial for combinatorial background	+0.27	+0.004
Total variation due to fit model	+0.27 -0.02	+0.011 -0.007

(plots in backup)

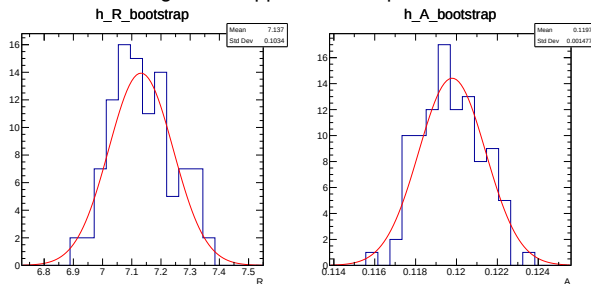
- small impact on A , since the fit models are the same for Λ_b^0 , $\bar{\Lambda}_b^0$ in the SUP decay
- moderate impact on R , since fit models for FAV and SUP decays are different

Systematic uncertainties due to efficiency corrections

Using alternative parameters of KDE in Meerkat (1–3%)



Cross check: using bootstrapped MC samples



Other systematic uncertainties

- PID: kernel width variation in the KDE procedure, and the uncertainty due to finite sample size of the calibration samples which is estimated as the RMS of bootstrapped samples (1–2%)
- Trigger efficiency: using single-track TOS efficiencies by CALO objects group ($\lesssim 0.5\%$)
- Charmless background: estimated from MC 0.5% and 1% in FAV and SUP, respectively
- Double misID background: estimated from MC 0.6% in FAV
- Single misID background: estimated from MC 0.01% and 0.1% in FAV and SUP, respectively
- Λ_b^0 , $\bar{\Lambda}_b^0$ production and detection asymmetry affecting A: taken from LHCb-ANA-2018-001 and L. Dufour thesis
 - ▶ Λ_b^0 , $\bar{\Lambda}_b^0$ production: 1.5%
 - ▶ p and $\bar{p}$: 1.5%
 - ▶ π^+ and π^- : 0.5%

Further checks (backup):

- measurement split by run period and magnet polarity
- fit range variation
- asymmetry of Λ_b^0 production and detection checked in favoured mode
- double misID background checked using $\Xi_b \rightarrow DpK$ decay

Systematic uncertainties in full phase space: summary

	R	A
Central value	7.14	0.119
Central value w/o efficiency correction	5.97	0.101
Statistical uncertainty	± 0.79	± 0.088
Systematic uncertainties		
fit model	+0.37 -0.15	+0.000 -0.011
efficiency corrections	+0.21 -0.24	+0.010 -0.008
PID	+0.08 -0.16	+0.001 -0.002
charmless background	± 0.08	
double misID background	± 0.005	
single misID background	± 0.001	
L0 TOS trigger efficiency	± 0.03	± 0.001
Λ_b^0 production asymmetry		± 0.015
p detection asymmetry		± 0.015
π detection asymmetry		± 0.005
Total systematic uncertainty	+0.43 -0.33	+0.024 -0.026

Systematic uncertainties in restricted phase space: summary

	$R_{M^2(pK^-) < 5 \text{ GeV}^2/c^4}$	$A_{M^2(pK^-) < 5 \text{ GeV}^2/c^4}$
Central value	8.55	0.007
Statistical uncertainty	± 1.47	± 0.158
Systematic uncertainties		
fit model	+0.27 -0.02	+0.011 -0.007
efficiency corrections	+0.25 -0.28	+0.010 -0.008
PID	+0.09 -0.19	+0.001 -0.002
charmless background	± 0.10	
double misID background	± 0.006	
single misID background	± 0.001	
L0 TOS trigger efficiency	± 0.03	± 0.001
Λ_b^0 production asymmetry		± 0.015
p detection asymmetry		± 0.015
π detection asymmetry		± 0.005
Total systematic uncertainty	+0.38 -0.33	+0.026 -0.024

Summary

Analysis of $\Lambda_b^0 \rightarrow [K^- \pi^+]_D p K^-$, $\Lambda_b^0 \rightarrow [K^+ \pi^-]_D p K^-$ using 9 fb^{-1} of Run 1+2 data:

- suppressed decay $\Lambda_b^0 \rightarrow [K^+ \pi^-]_D p K^-$ observed for the first time
- measured ratio of branching fractions and asymmetry:

- ▶ in the full phase space:

$$R = \frac{\mathcal{B}(\Lambda_b^0 \rightarrow [K^- \pi^+]_D p K^-)}{\mathcal{B}(\Lambda_b^0 \rightarrow [K^+ \pi^-]_D p K^-)} = 7.14 \pm 0.79(\text{stat.})_{-0.33}^{+0.43}(\text{syst.})$$

$$A = \frac{N_{\text{SUP}}(\Lambda_b^0) - N_{\text{SUP}}(\overline{\Lambda_b^0})}{N_{\text{SUP}}(\Lambda_b^0) + N_{\text{SUP}}(\overline{\Lambda_b^0})} = 0.119 \pm 0.088(\text{stat.})_{-0.026}^{+0.024}(\text{syst.})$$

- ▶ in restricted phase space $M^2(pK) < 5 \text{ GeV}^2$ with expected enhanced sensitivity to γ :

$$R = 8.55 \pm 1.47(\text{stat.})_{-0.33}^{+0.38}(\text{syst.})$$

$$A = 0.01 \pm 0.16(\text{stat.})_{-0.024}^{+0.026}(\text{syst.})$$

- Many thanks to:

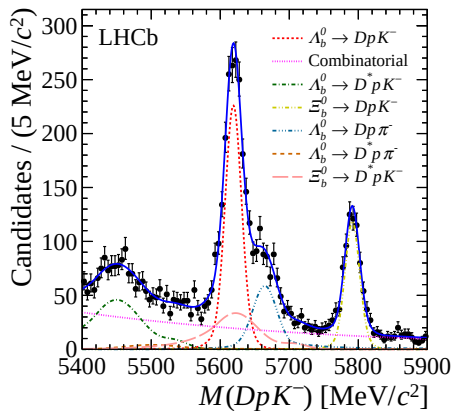
- ▶ RC Steve Blusk and Fabio Ferrari
- ▶ EB reviewer Franz Muheim
- ▶ B2OC WG conveners Nathan Jurik, Paras Naik and WG reader Nicole Skidmore
- ▶ all who provided comments to this analysis

- **Seek approval to go to paper**

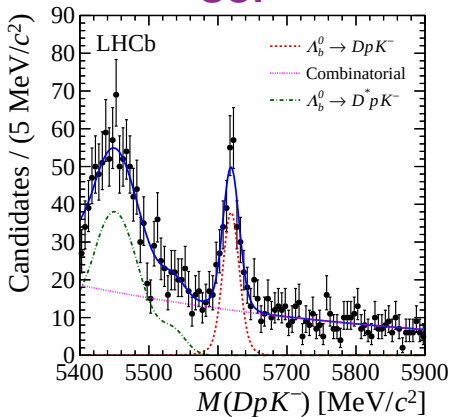
BACKUP

Invariant mass fit (paper plots)

FAV

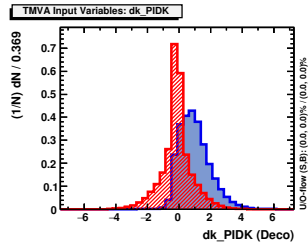
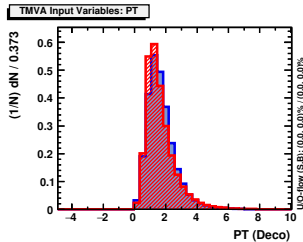
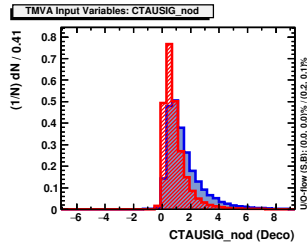
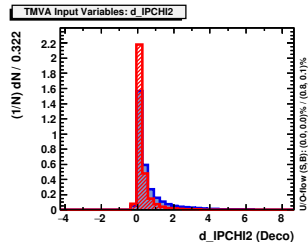
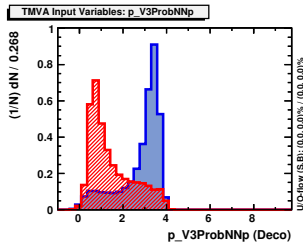
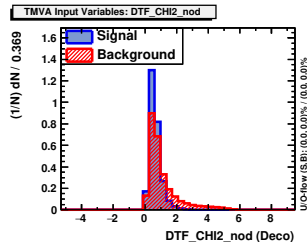


SUP

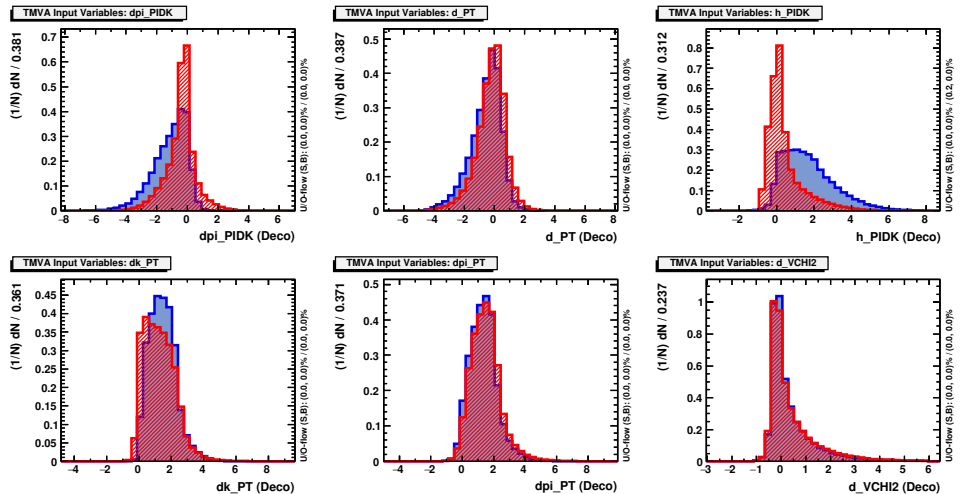


	Number of generated events					
	2011	2012	2015	2016	2017	2018
$\Lambda_b^0 \rightarrow (D^0 \rightarrow K^- \pi^+) p K^-$	607,271	1,060,536	307,874	501,197	505,078	765,058
$\Lambda_b^0 \rightarrow (D^0 \rightarrow K^- K^+) p K^-$	n/u	n/u	n/u	1,208,421	n/u	n/u
$\Lambda_b^0 \rightarrow (D^0 \rightarrow \pi^- \pi^+) p K^-$	n/u	n/u	n/u	1,200,540	n/u	n/u
$\Lambda_b^0 \rightarrow (D^0 \rightarrow K^- \pi^+) p \pi^-$	529,646	1,003,900	315,029	505,150	513,105	507,737
$\Lambda_b^0 \rightarrow D^{*0} p K^-$	n/a	2,019,493	n/a	n/a	1,903,870	1,018,057
$\Lambda_b^0 \rightarrow D^{*0} p \pi^-$	n/a	2,047,991	n/a	n/a	2,019,514	1,024,366

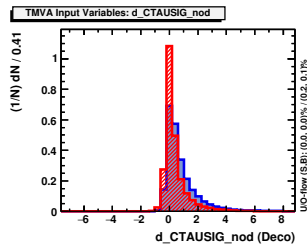
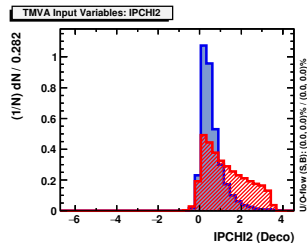
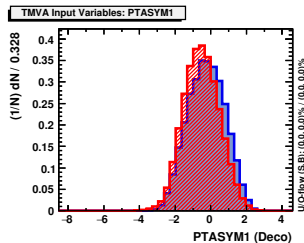
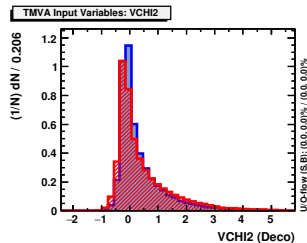
BDT: distributions of input variables (1)



BDT: distributions of input variables (2)

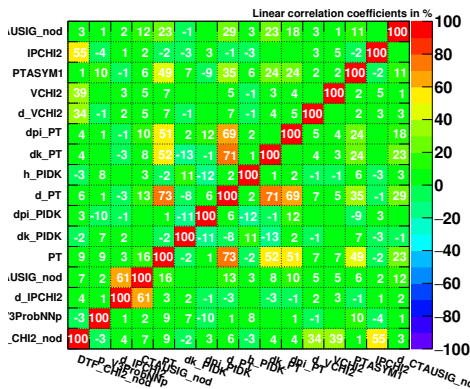


BDT: distributions of input variables (3)

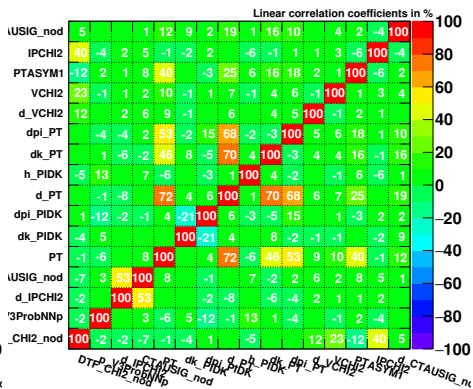


BDT: correlations of input variables

Correlation Matrix (signal)



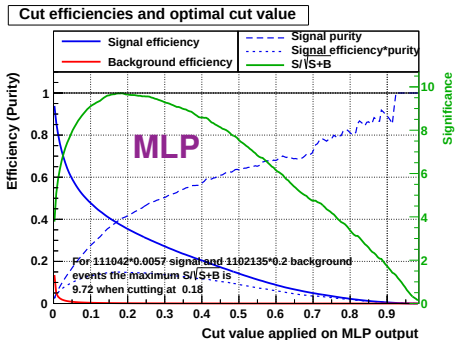
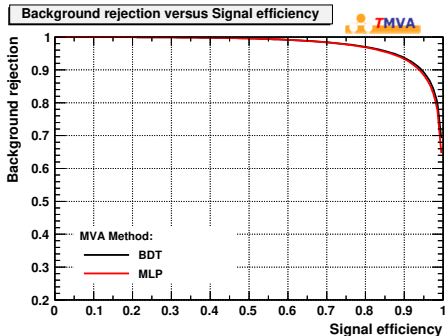
Correlation Matrix (background)



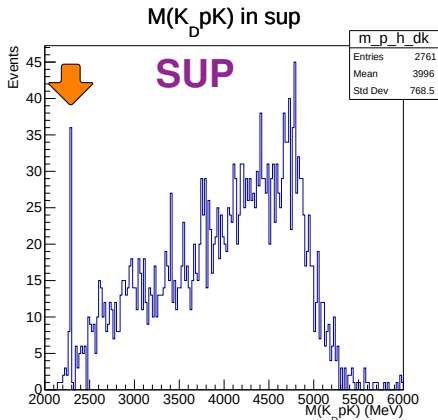
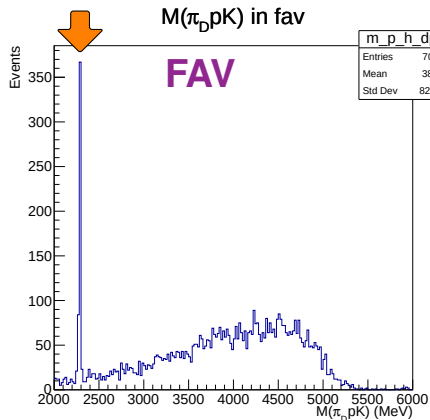
BDT: ranking of input variables

: Rank	: Variable	: Variable Importance
: 1	: dk_PIDK	: 1.526e-01
: 2	: h_PIDK	: 1.160e-01
: 3	: p_V3ProbNNp	: 1.054e-01
: 4	: DTF_CHI2_nod	: 9.894e-02
: 5	: dpi_PIDK	: 9.556e-02
: 6	: CTAUSIG_nod	: 8.181e-02
: 7	: IPCHI2	: 6.543e-02
: 8	: dpi_PT	: 4.831e-02
: 9	: d_CTAUSIG_nod	: 4.802e-02
: 10	: d_PT	: 3.878e-02
: 11	: PTASYM1	: 3.672e-02
: 12	: PT	: 3.018e-02
: 13	: dk_PT	: 2.785e-02
: 14	: d_VCHI2	: 2.234e-02
: 15	: VCHI2	: 1.859e-02
: 16	: d_IPCHI2	: 1.345e-02

BDT: comparison vs MLP

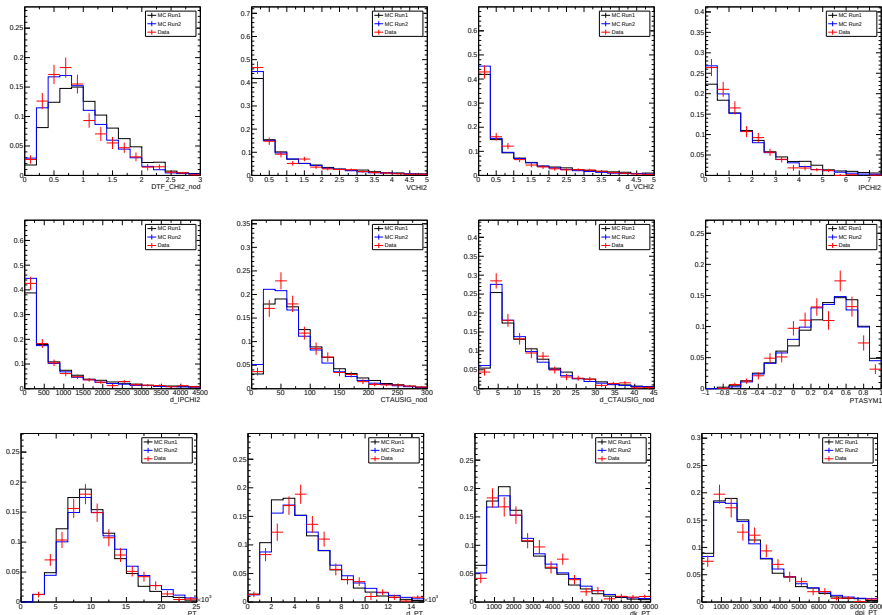


$$L_c^+ \rightarrow ph^- h^+ \text{ veto}$$

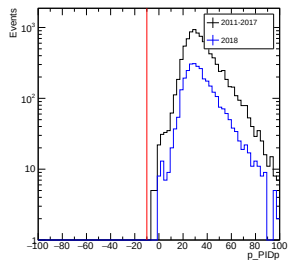
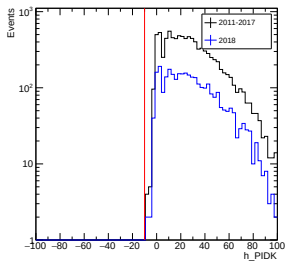
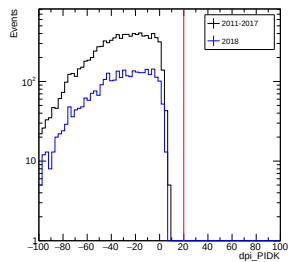
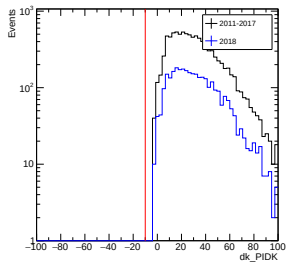


- $L_c^+ \rightarrow ph^- h^+$ contributes to FAV (SUP) when h^+ is $\pi(K)$ from D^0 and h^- is K from Λ_D^0 (if h^-, h^+ are both from D^0 , $M(ph^- h^+)$ is always greater than $M(L_c^+)$)
- suppress by vetoing $2270 < M(ph^- h^+) < 2305$ MeV

Topological variables in MC and data (s-weighted)



PID variables used in stripping

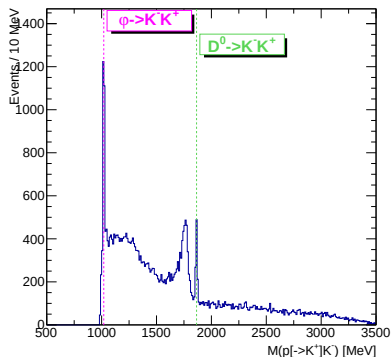


Other potential backgrounds checked in data (SUP)

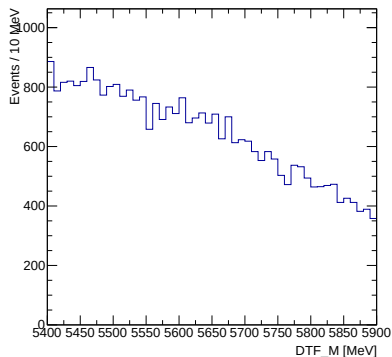
Loose event selection and inverted PID requirement $p_{PIDK} > 15$:

- contribution from partially reconstructed decays of D^{*0} and D_s^{*0} is seen
- no peaking structure in $M(DpK)$

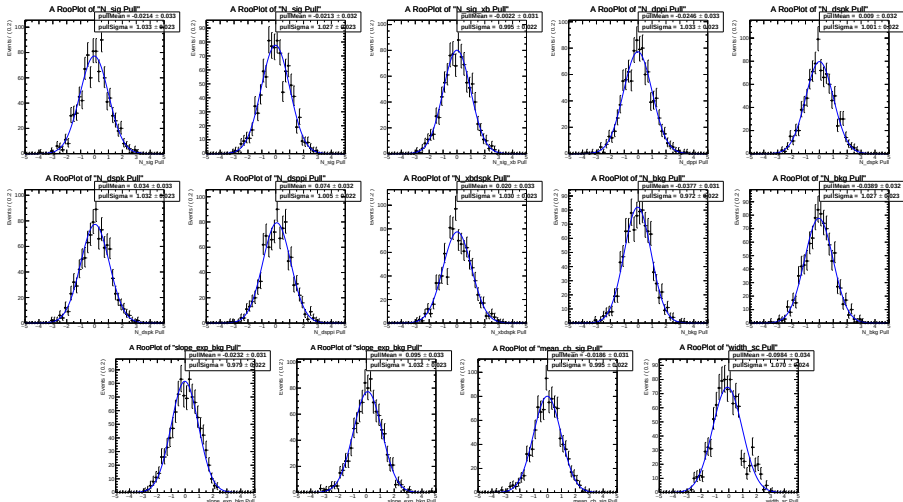
SUP



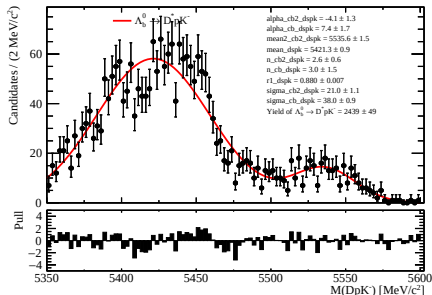
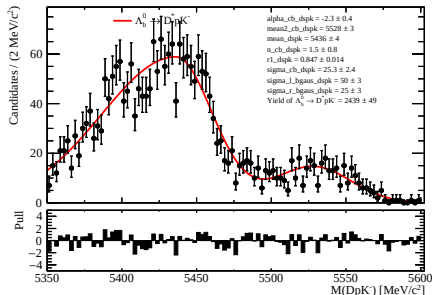
SUP



Validation of fitting procedure using toys

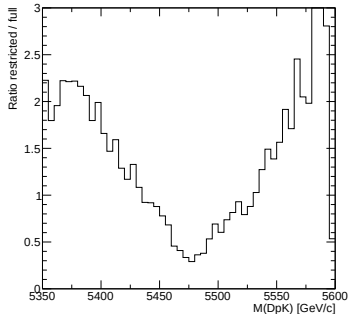
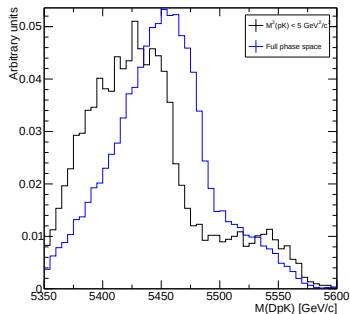


$\Lambda_b^0 \rightarrow D^* p K$ in $M^2(pK) < 5 \text{ GeV}^2$



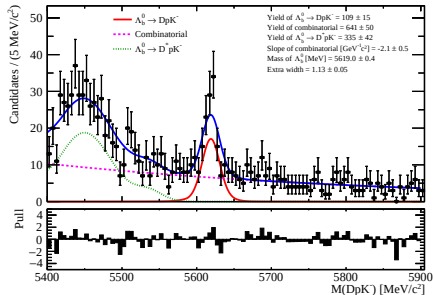
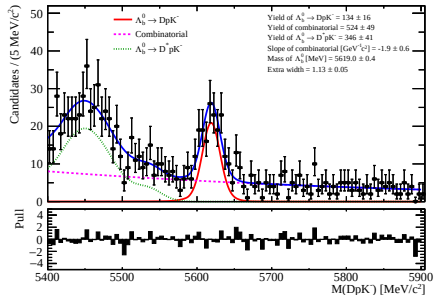
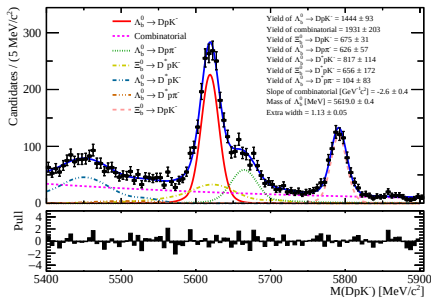
- Left: nominal shape (bifurcated Gaus + CB)
- Right: variation (CB + CB) used for systematics

$$\Lambda_b^0 \rightarrow D^* p K \text{ in } M^2(pK) < 5 \text{ GeV}^2$$

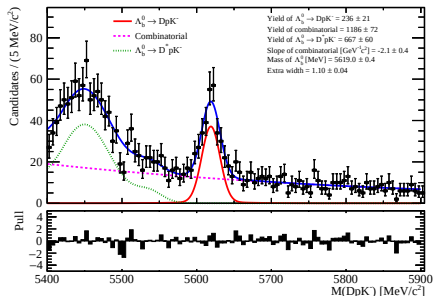
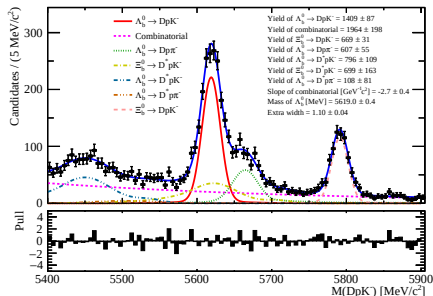
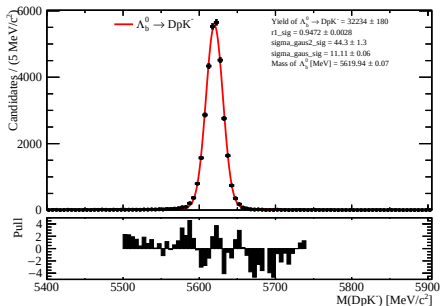


- $M(DpK)$ invariant mass can take values from $m(D) + M(pK)$ to $m(\Lambda_b^0) - m(\pi^0)$ (for $D^{*0} \rightarrow D^0 \pi^0$), or to $m(\Lambda_b^0)$ (for $D^{*0} \rightarrow D^0 \gamma$)
(the lower boundary depends on $M(pK)$, but the upper boundary does not depend on it)
- summing up over the full range of $M(pK)$ gives a distribution which is shifted to higher values than when summing over the restricted $M^2(pK) < 5 \text{ GeV}^2$ range

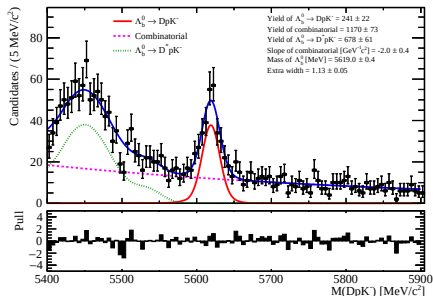
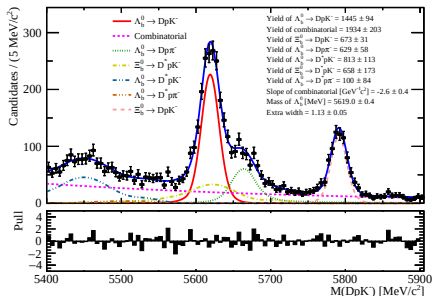
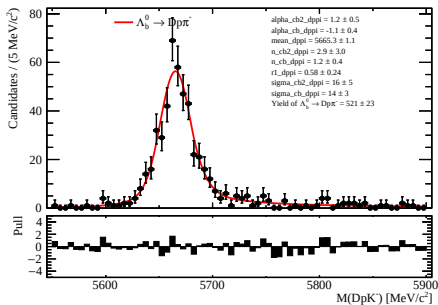
Fits to FAV, Λ_b^0 SUP and $\bar{\Lambda}_b^0$ SUP data samples



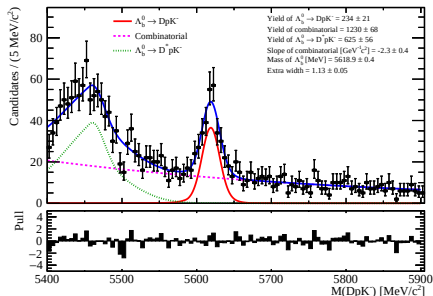
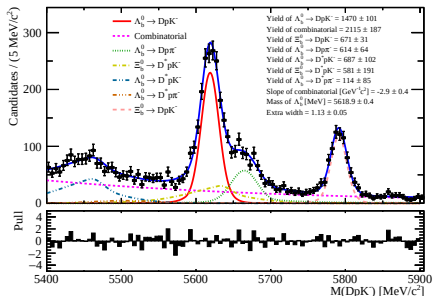
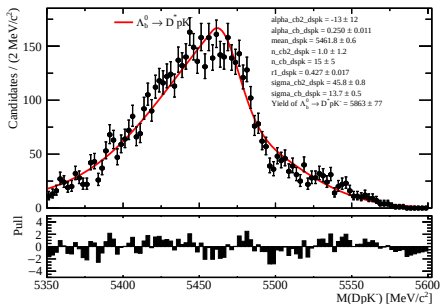
Fits using alternative $\Lambda_b^0 \rightarrow D^0 p K^-$ shape (Gaus+Gaus)



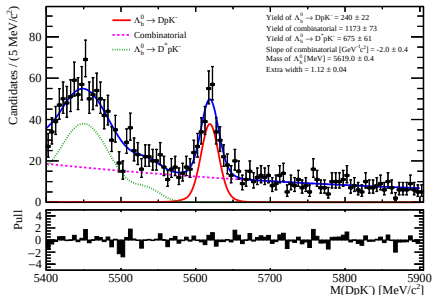
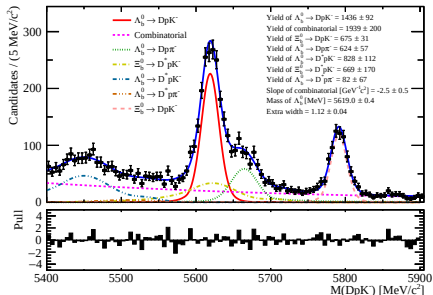
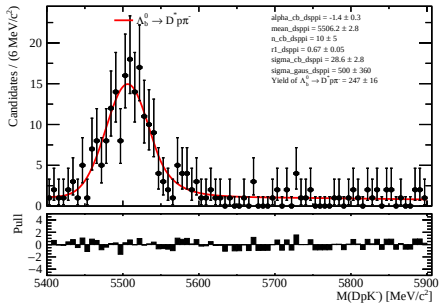
Fits using alternative $\Lambda_b^0 \rightarrow D^0 p \pi^-$ shape (CB+CB)



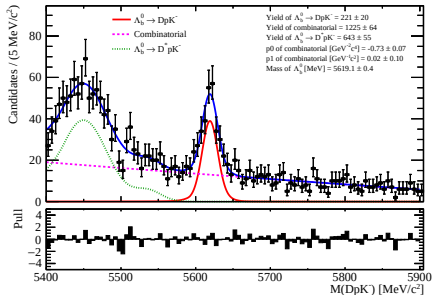
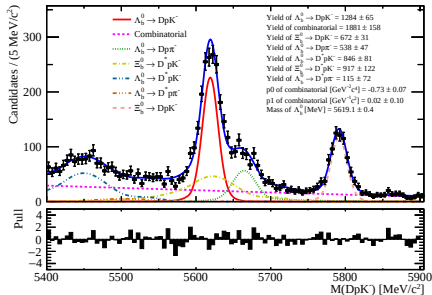
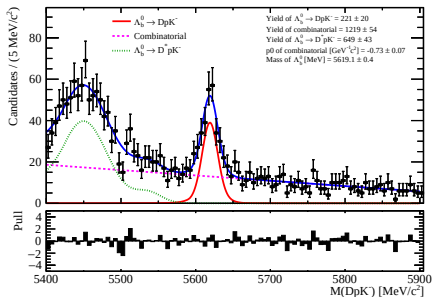
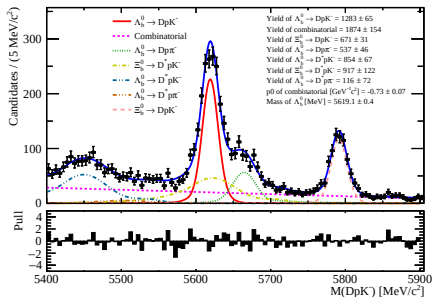
Fits using alternative $\Lambda_b^0 \rightarrow D^{*0} p K^-$ shape (CB+CB with shared mean)



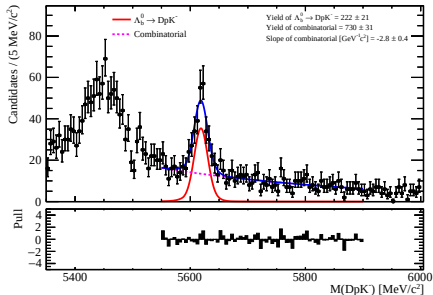
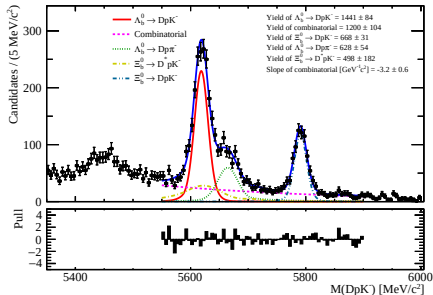
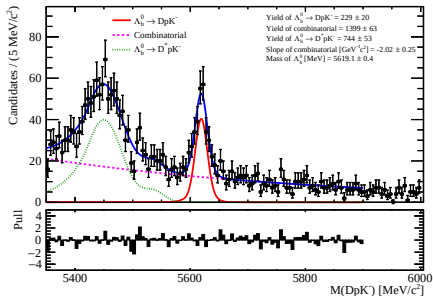
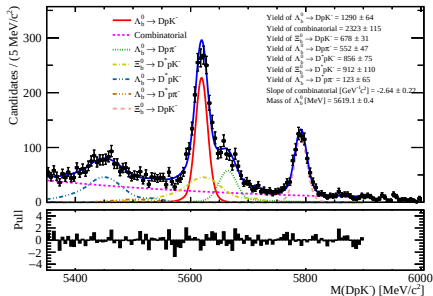
Fits using alternative $\Lambda_b^0 \rightarrow D^{*0} p \pi^-$ shape (CB+Gaus)



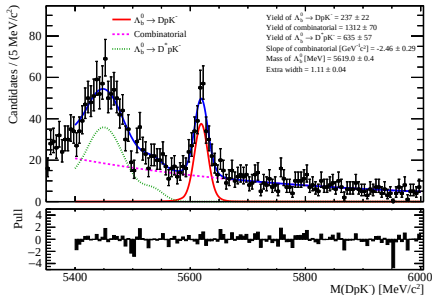
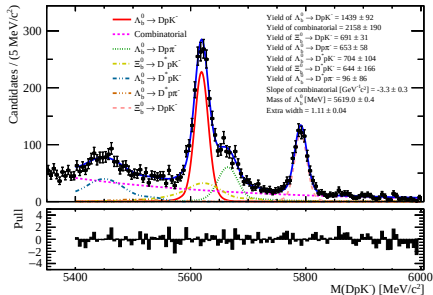
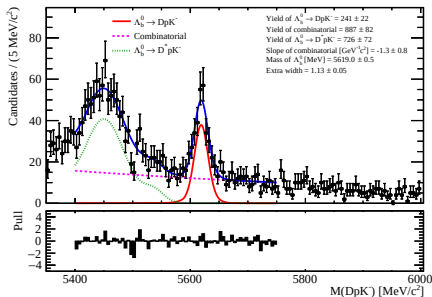
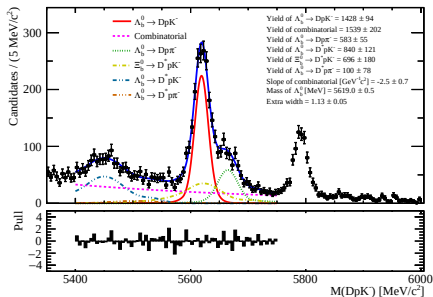
Fits using alternative comb. bkg. shape (1st and 2nd order polynomials)



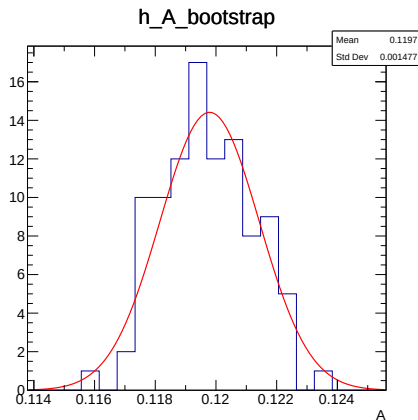
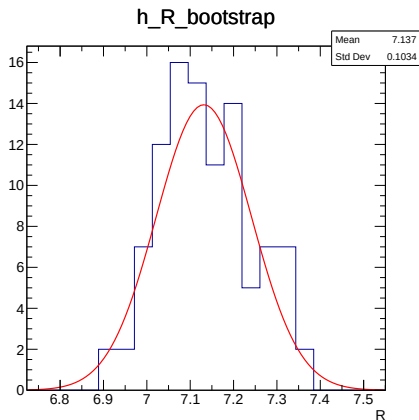
Fits using alternative fit ranges (1)



Fits using alternative fit ranges (2)



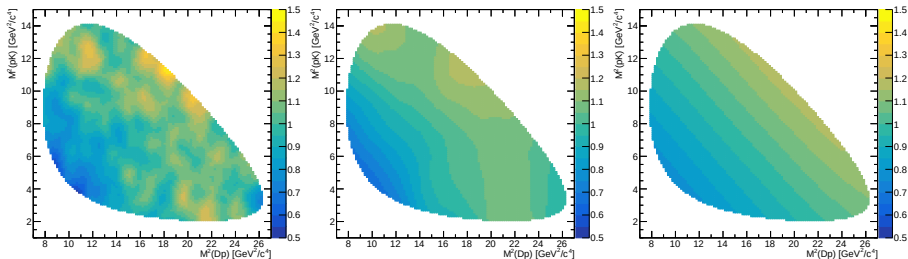
MC bootstrapping



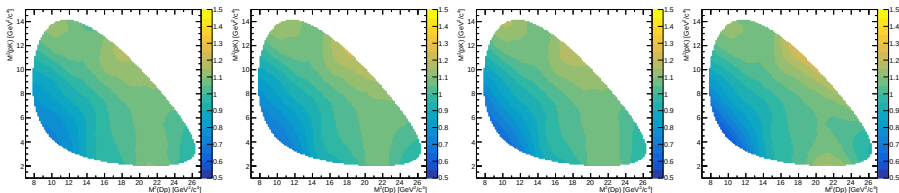
- standard deviations are 1% and 0.1% for R and A , respectively: within already assigned systematic uncertainties on efficiency corrections

Systematic uncertainties on efficiency corrections (1)

Relative efficiency corrections determined using kernel width of 1, 3 (nominal) and 10 GeV^2 (left to right):

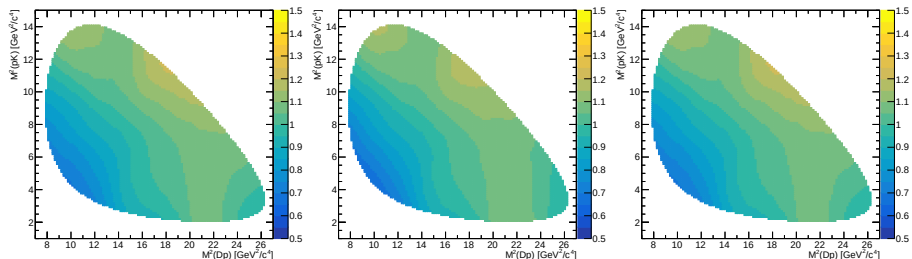


Relative efficiency corrections determined using no approximation PDF, a first-order polynomial, a second-order polynomial (nominal), and a third-order polynomial as approximation PDF (left to right):

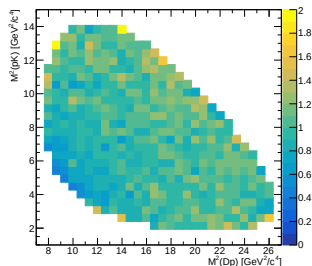


Systematic uncertainties on efficiency corrections (2)

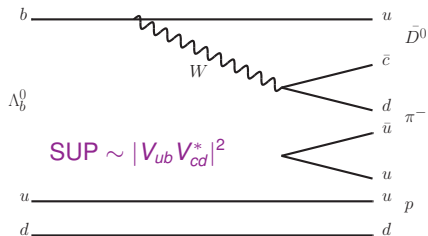
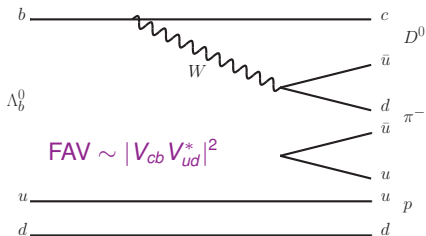
Relative efficiency corrections determined using 50, 200 (nominal) and 500 bins (left to right):



Relative efficiency corrections determined in coarse bins of $M^2(D^0p)$ and $M^2(pK^-)$ w/o KDE:



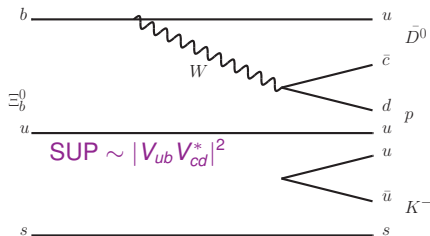
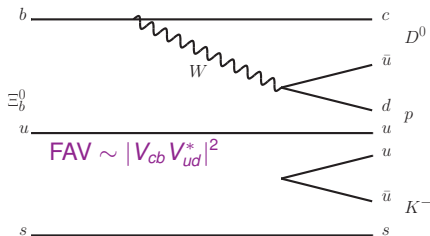
$$\Lambda_b^0 \rightarrow [K^- \pi^+]_{D^0} p \pi^-, \Lambda_b^0 \rightarrow [K^+ \pi^-]_{D^0} p \pi^-$$



- for $\Xi_b \rightarrow D^0 p K$, $\left| \frac{V_{cb} V_{ud}^*}{V_{ub} V_{cd}^*} \right|^2 \approx 2580$

- $\frac{\mathcal{B}(D^0 \rightarrow K^- \pi^+)}{\mathcal{B}(D^0 \rightarrow K^+ \pi^-)} \approx 263$

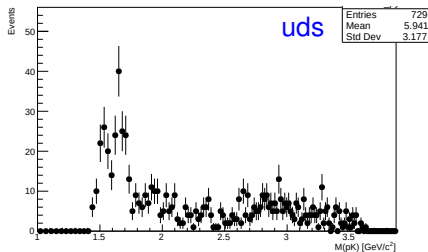
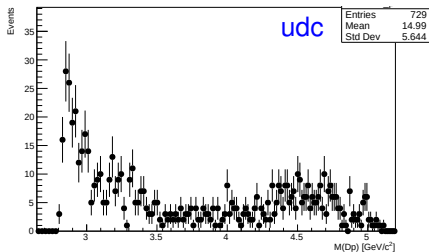
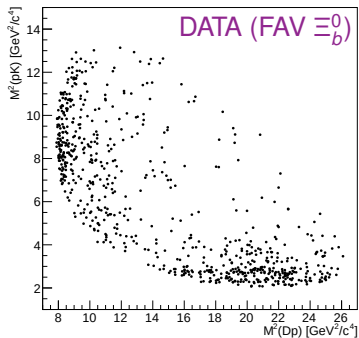
$$\Xi_b^0 \rightarrow DKp$$



- for $\Xi_b \rightarrow D^0 p K$, $\left| \frac{V_{cb} V_{ud}^*}{V_{ub} V_{cd}^*} \right|^2 \approx 2580$

- $\frac{\mathcal{B}(D^0 \rightarrow K^- \pi^+)}{\mathcal{B}(D^0 \rightarrow K^+ \pi^-)} \approx 263$

Dalitz plot $\Xi_b^0 \rightarrow D^0 K p$



R and A obtained from data split by run period or magnet polarity

Sample	R	A
run1	6.71 ± 1.90	0.14 ± 0.21
run2	7.17 ± 0.87	0.11 ± 0.10
magdown	8.08 ± 1.18	0.25 ± 0.12
magup	6.13 ± 0.95	0.00 ± 0.12

(statistical uncertainties only)